Restricted-Domain Computational and Geometric Investigation of
$a^3 + b^3 + c^3 = d^3$

Full exhaustive enumeration, parametric classification, near-miss analysis, symbolic verification, and multi-dimensional visualization over $1 \le a \le b \le c < d \le N$, with $N = 500$.

Kepler Agent — deep-research report · generated 2026-08-04 · all computation performed in a Daytona sandbox (exact integer arithmetic).

1. Executive summary

$$a^3 + b^3 + c^3 = d^3, \quad 1 \le a \le b \le c < d \le N = 500$$

We exhaustively searched the bounded lattice box `1 ≤ a ≤ b ≤ c < d ≤ 500` for integer solutions of the sum-of-three-cubes equation. The search is **complete and exact**: every candidate triple `(a,b,c)` was examined once using 64-bit integer arithmetic, and membership of the target `a³+b³+c³` in the set of cubes was decided by binary search — no floating-point rounding enters the solution test.

889
total solutions (a≤b≤c<d≤500)

338
primitive solutions (gcd = 1)

20,833,250
triples examined (exhaustive)

83
in family A: multiples of (3,4,5,6)

138
in family B: Ramanujan (m,n)

668
sporadic (outside A ∪ B)

Key findings:

- **Control recovered.** The classical identity $3^3+4^3+5^3=6^3$ and **all 83 of its multiples** `(3k,4k,5k,6k)` with $6k \le 500$ were recovered and flagged automatically.
- **Two parametric families identified.** Family A (multiples of the base (3,4,5,6)) and Family B (Ramanujan's two-parameter cubic identity) together account for 221 of the 889 solutions; the remaining 668 are **sporadic** within this box.
- **Both identities symbolically proved.** SymPy expansion gives residual exactly 0 for both the (3,4,5,6)·k identity and the Ramanujan family, corroborated by 50,000 random large-integer numeric checks with 0 failures.
- **Density.** Solutions become steadily more common with d: the count grows roughly like a smooth polynomial in N (see runtime/scaling analysis), and the proportion primitive settles near 38.0%.
- **Near-misses.** For most sampled d a genuine solution exists ($\delta(d)=0$); where none does, the minimal residual is tiny relative to d^3 (see the near-miss landscape).

Scope & honesty. Every statement below is a statement about the finite box N=500. Nothing here bears on the open questions surrounding sums of three cubes in general (e.g. which integers are sums of three cubes over $\mathbb{Z}$); this is a rigorous *finite* census, not a theorem about all integers.

2. Computational enumeration & data tables

2.1 Algorithm

For each ordered pair `(a,b)` with `a ≤ b` we form the vector `s = a³ + b³ + c³` over all `c ∈ [b, N-1]` and test, by a vectorised binary search (`numpy.searchsorted`) into the sorted array of cubes `{1³,…,N³}` , whether `s` equals some `d³` with `d ≤ N` . Because `a,b ≥ 1` we always have `d > c` , so the constraint `c < d` is automatic. The test is exact integer arithmetic and each candidate triple is examined **exactly once**; total triples examined = **20,833,250**, matching the closed-form count $\sum_{a\le b}(N\text{-}b)$.

1.45s
enumeration wall-clock (N=500)

5.67s
near-miss sweep wall-clock

66 MB
peak resident memory

2.2 Primitive solutions (complete list, d ≤ 500)

All 338 primitive solutions (gcd(a,b,c,d)=1), sorted by d. The ten smallest are (3,4,5,6); (1,6,8,9); (3,10,18,19); (7,14,17,20); (4,17,22,25); (18,19,21,28); (11,15,27,29); (2,17,40,41); (6,32,33,41); (16,23,41,44).

Table 1. Complete list of primitive solutions with d ≤ 500 (columns: a, b, c, d, parametric family).

a	b	c	d	family
3	4	5	6	A:(3,4,5,6)-multiple
1	6	8	9	sporadic
3	10	18	19	sporadic
7	14	17	20	B:Ramanujan(m,n)
4	17	22	25	sporadic
18	19	21	28	B:Ramanujan(m,n)
11	15	27	29	sporadic
2	17	40	41	sporadic
6	32	33	41	sporadic
16	23	41	44	sporadic
3	36	37	46	B:Ramanujan(m,n)
27	30	37	46	B:Ramanujan(m,n)
29	34	44	53	sporadic
12	19	53	54	sporadic
15	42	49	58	B:Ramanujan(m,n)
22	51	54	67	sporadic
36	38	61	69	sporadic
7	54	57	70	B:Ramanujan(m,n)
14	23	70	71	sporadic
34	39	65	72	sporadic
38	43	66	75	sporadic
31	33	72	76	sporadic

a	b	c	d	family
25	48	74	81	sporadic
19	60	69	82	B:Ramanujan(m,n)
28	53	75	84	sporadic
50	61	64	85	sporadic
20	54	79	87	sporadic
26	55	78	87	sporadic
38	48	79	87	sporadic
21	43	84	88	sporadic
25	31	86	88	sporadic
17	40	86	89	sporadic
25	38	87	90	sporadic
58	59	69	90	B:Ramanujan(m,n)
32	54	85	93	sporadic
19	53	90	96	sporadic
45	69	79	97	sporadic
12	31	102	103	sporadic
33	70	92	105	sporadic
13	51	104	108	sporadic
15	82	89	108	B:Ramanujan(m,n)
29	75	96	110	sporadic
16	47	108	111	sporadic
50	74	97	113	sporadic
3	34	114	115	sporadic
23	86	97	116	B:Ramanujan(m,n)
9	55	116	120	sporadic
49	84	102	121	sporadic
19	92	101	122	B:Ramanujan(m,n)
44	51	118	123	sporadic
23	94	105	126	B:Ramanujan(m,n)
13	65	121	127	sporadic
38	57	124	129	sporadic
5	76	123	132	sporadic
86	95	97	134	B:Ramanujan(m,n)
44	73	128	137	sporadic
94	96	99	139	sporadic
72	85	122	141	sporadic
31	64	137	142	sporadic
1	71	138	144	sporadic
12	81	136	145	sporadic
22	75	140	147	sporadic
71	73	138	150	sporadic
46	47	148	151	sporadic
65	87	142	156	sporadic
3	121	131	159	sporadic

a	b	c	d	family
69	123	124	160	sporadic
59	93	148	162	sporadic
69	99	146	164	sporadic
12	86	159	167	sporadic
96	107	141	170	sporadic
54	80	163	171	sporadic
107	108	136	171	sporadic
1	135	138	172	B:Ramanujan(m,n)
47	97	162	174	sporadic
25	92	167	176	sporadic
48	137	142	177	sporadic
48	133	147	178	sporadic
17	57	177	179	sporadic
108	109	150	181	sporadic
57	68	180	185	sporadic
68	113	166	185	sporadic
18	121	167	186	sporadic
58	131	160	187	sporadic
115	122	149	188	B:Ramanujan(m,n)
21	46	188	189	sporadic
56	133	163	190	sporadic
39	146	156	191	sporadic
34	123	173	192	sporadic
25	68	190	193	sporadic
53	58	194	197	sporadic
27	46	197	198	sporadic
73	135	170	198	sporadic
6	127	180	199	sporadic
45	53	199	201	sporadic
76	141	171	202	sporadic
85	138	171	202	B:Ramanujan(m,n)
81	147	167	203	sporadic
108	142	165	205	sporadic
5	163	164	206	B:Ramanujan(m,n)
35	77	202	206	sporadic
57	146	180	209	sporadic
113	146	166	209	sporadic
42	83	205	210	sporadic
56	61	210	213	sporadic
16	51	213	214	sporadic
58	157	179	214	sporadic
46	126	201	217	sporadic
93	155	180	218	sporadic
140	151	161	218	B:Ramanujan(m,n)

a	b	c	d	family
50	67	216	219	sporadic
51	152	190	219	sporadic
67	167	177	219	sporadic
108	163	170	219	sporadic
31	95	219	225	sporadic
18	167	193	228	sporadic
76	165	192	229	sporadic
102	157	192	229	sporadic
56	102	223	231	sporadic
98	136	207	231	sporadic
103	140	204	231	sporadic
85	107	220	232	sporadic
123	125	208	234	sporadic
54	163	204	235	sporadic
78	172	195	235	sporadic
96	163	198	235	sporadic
147	157	186	238	sporadic
15	114	230	239	sporadic
23	178	200	239	sporadic
113	124	220	241	sporadic
73	174	207	244	B:Ramanujan(m,n)
113	166	207	246	B:Ramanujan(m,n)
4	57	248	249	sporadic
6	179	216	251	sporadic
163	164	197	254	B:Ramanujan(m,n)
9	58	255	256	sporadic
9	183	220	256	sporadic
22	57	255	256	sporadic
101	178	219	258	B:Ramanujan(m,n)
65	127	248	260	sporadic
19	93	258	262	sporadic
49	80	263	266	sporadic
66	117	260	269	sporadic
152	158	229	269	sporadic
23	102	265	270	sporadic
158	171	228	275	sporadic
71	150	262	279	sporadic
138	200	223	279	sporadic
50	172	257	281	sporadic
81	202	239	282	B:Ramanujan(m,n)
113	116	271	284	sporadic
95	196	248	287	sporadic
71	177	262	288	sporadic
57	126	280	289	sporadic

a	b	c	d	family
179	188	229	290	B:Ramanujan(m,n)
58	201	255	292	sporadic
106	227	229	292	sporadic
183	201	220	292	sporadic
8	229	236	293	sporadic
14	189	264	293	sporadic
159	214	228	295	sporadic
47	75	295	297	sporadic
15	64	297	298	sporadic
17	191	274	302	sporadic
23	81	300	302	sporadic
55	113	296	302	sporadic
145	179	267	303	sporadic
151	162	272	303	sporadic
53	153	291	305	sporadic
27	64	306	307	sporadic
7	119	303	309	sporadic
50	184	285	309	sporadic
93	242	244	309	sporadic
102	194	279	311	sporadic
15	213	281	317	sporadic
162	173	282	317	sporadic
73	120	314	321	sporadic
105	196	290	321	sporadic
152	219	270	323	sporadic
41	114	319	324	sporadic
102	227	277	324	sporadic
173	214	267	324	B:Ramanujan(m,n)
107	230	277	326	B:Ramanujan(m,n)
31	248	270	327	sporadic
44	199	300	327	sporadic
135	164	304	327	sporadic
144	227	277	330	sporadic
88	222	291	331	sporadic
185	218	271	332	B:Ramanujan(m,n)
42	181	315	334	sporadic
183	192	290	335	sporadic
40	203	309	336	sporadic
3	214	309	340	sporadic
192	249	263	344	sporadic
5	232	307	346	sporadic
48	268	285	349	sporadic
36	147	341	350	sporadic
174	247	284	351	sporadic

a	b	c	d	family
127	204	321	352	sporadic
158	241	295	352	sporadic
27	232	317	354	sporadic
100	213	323	354	sporadic
12	238	315	355	sporadic
106	185	335	356	sporadic
174	239	297	356	sporadic
227	230	277	356	B:Ramanujan(m,n)
118	225	324	361	sporadic
20	151	353	362	sporadic
95	229	327	363	sporadic
40	273	303	364	sporadic
107	209	337	365	sporadic
116	271	305	368	sporadic
134	151	353	368	sporadic
121	122	360	369	sporadic
186	236	313	369	sporadic
10	261	323	372	sporadic
99	200	349	372	sporadic
40	141	366	373	sporadic
72	257	327	374	sporadic
3	245	340	378	sporadic
119	268	321	378	B:Ramanujan(m,n)
123	276	316	379	sporadic
242	262	285	381	sporadic
20	109	379	382	sporadic
160	191	356	383	sporadic
99	236	349	384	sporadic
173	232	339	384	sporadic
185	207	346	384	sporadic
168	268	321	385	sporadic
71	81	384	386	sporadic
87	297	316	388	sporadic
138	145	375	388	sporadic
141	272	330	389	sporadic
24	159	382	391	sporadic
87	150	382	391	sporadic
17	175	380	392	sporadic
79	245	357	393	sporadic
135	177	376	394	sporadic
147	232	357	394	sporadic
54	152	387	395	sporadic
24	197	379	396	sporadic
94	149	387	396	sporadic

a	b	c	d	family
240	271	305	396	sporadic
38	295	337	400	sporadic
114	261	355	400	sporadic
154	251	357	402	sporadic
177	276	343	406	B:Ramanujan(m,n)
178	295	329	406	sporadic
32	310	335	407	sporadic
64	107	405	408	sporadic
123	197	388	408	sporadic
149	256	363	408	sporadic
180	197	379	408	sporadic
108	267	365	410	sporadic
80	219	388	411	sporadic
171	282	349	412	B:Ramanujan(m,n)
88	95	412	415	sporadic
18	137	412	417	sporadic
169	234	380	417	sporadic
107	292	362	419	sporadic
235	236	362	419	sporadic
144	179	405	422	sporadic
138	178	407	423	sporadic
22	320	353	425	sporadic
178	188	401	425	sporadic
149	166	411	426	sporadic
160	243	389	426	sporadic
191	290	361	428	B:Ramanujan(m,n)
276	303	313	430	B:Ramanujan(m,n)
135	188	414	431	sporadic
200	239	388	431	sporadic
213	323	334	432	sporadic
219	269	370	432	sporadic
73	123	431	435	sporadic
18	193	423	436	sporadic
104	150	429	437	sporadic
131	261	400	438	sporadic
107	306	381	440	sporadic
202	261	396	445	sporadic
215	218	409	446	sporadic
127	188	432	447	sporadic
138	239	418	447	sporadic
5	216	436	453	sporadic
26	312	397	453	sporadic
72	157	446	453	sporadic
234	260	397	453	sporadic

a	b	c	d	family
58	75	453	454	sporadic
69	328	387	454	sporadic
283	294	357	454	B:Ramanujan(m,n)
19	362	365	458	B:Ramanujan(m,n)
30	266	427	459	sporadic
97	333	389	459	sporadic
160	326	387	459	sporadic
123	136	453	460	sporadic
5	86	460	461	sporadic
131	212	445	464	sporadic
49	122	462	465	sporadic
142	362	369	465	sporadic
60	342	395	467	sporadic
100	242	447	471	sporadic
191	354	376	471	sporadic
97	320	415	472	sporadic
111	148	465	472	sporadic
85	353	397	475	sporadic
115	366	384	475	sporadic
89	231	456	476	sporadic
219	223	443	477	sporadic
51	82	477	478	sporadic
64	75	477	478	sporadic
47	350	406	479	sporadic
109	293	437	479	sporadic
256	311	398	479	sporadic
51	232	461	480	sporadic
139	309	428	480	sporadic
233	279	424	480	sporadic
191	332	409	482	B:Ramanujan(m,n)
109	170	475	484	sporadic
179	267	447	485	sporadic
54	183	478	487	sporadic
20	89	487	488	sporadic
193	276	446	489	sporadic
268	362	369	489	sporadic
75	366	410	491	sporadic
48	85	491	492	sporadic
113	264	463	492	sporadic
149	336	427	492	sporadic
190	279	449	492	sporadic
243	358	389	492	sporadic
281	322	399	492	B:Ramanujan(m,n)
18	349	426	493	sporadic

a	b	c	d	family
118	160	485	493	sporadic
260	369	375	494	sporadic
254	327	412	495	sporadic
57	82	495	496	sporadic
315	322	389	498	B:Ramanujan(m,n)
123	168	490	499	sporadic
42	329	447	500	sporadic
82	119	497	500	sporadic

2.3 Summary statistics

Table 2. Solutions per d-bin: total count, primitive count, proportion primitive, and min / max / mean of the ratio d / max(a,b,c).

d_bin	n_solutions	n_primitive	prop_primitive	min_ratio_d_over_maxabc	max_ratio_d_over_maxabc	mean_ratio
1-100	82	37	0.4512	1.0143	1.3333	1.1507
101-200	144	57	0.3958	1.0051	1.404	1.1412
201-300	190	69	0.3632	1.0034	1.404	1.1412
301-400	218	82	0.3761	1.0033	1.3368	1.1313
401-500	255	93	0.3647	1.002	1.404	1.134
ALL	889	338	0.3802	1.002	1.404	1.1376

2.4 Runtime & pair-count audit

Table 5. Exact number of triples / evaluations examined, wall-clock time, and peak memory for the enumeration, the near-miss sweep, and the N-scaling runs.

stage	N	triples_or_evals_examined	wall_time_s	peak_mem_MB
enumeration	500	20833250	1.45	65.6
near_miss	500	62499750	5.673	65.6
enum_scale	100	166650	0.047	
enum_scale	200	1333300	0.199	
enum_scale	300	4499950	0.475	
enum_scale	400	10666600	0.902	
enum_scale	500	20833250	1.433	

3. Parametric classification & symbolic verification

3.1 The two classical families

Family A — multiples of (3,4,5,6). The seed $3^3+4^3+5^3=6^3$ scales: `(3k)³+(4k)³+(5k)³=(6k)³` for every integer $k \geq 1$.

Family B — Ramanujan's two-parameter identity. With integer parameters (m,n),

$$a = 3m^2+5mn-5n^2,\ \ b = 4m^2-4mn+6n^2,\ \ c = 5m^2-5mn-3n^2,\ \ d = 6m^2-4mn+4n^2$$

satisfies $a^3+b^3+c^3=d^3$ identically. Enumerating (m,n) and reducing to primitive form yields 1,079 distinct primitive Ramanujan tuples overall; 138 of the in-box solutions (and their scalings) fall in this family.

3.2 Membership of the enumerated solutions

Table 4. Parametric-family membership: counts, primitive counts, and representative examples for family A, family B, and the sporadic remainder.

family	count	n_primitive	examples
A:(3,4,5,6)-multiple	83	1	(3, 4, 5, 6); (6, 8, 10, 12); (9, 12, 15, 18); (12, 16, 20, 24); (15, 20, 25, 30)
B:Ramanujan(m,n)	138	38	(7, 14, 17, 20); (18, 19, 21, 28); (14, 28, 34, 40); (3, 36, 37, 46); (27, 30, 37, 46)
sporadic	668	299	(1, 6, 8, 9); (2, 12, 16, 18); (3, 10, 18, 19); (4, 17, 22, 25); (3, 18, 24, 27)

3.3 Symbolic + numeric verification

SymPy expansion (exact):
$(3k)^3+(4k)^3+(5k)^3-(6k)^3$ → `0`
$a^3+b^3+c^3-d^3$ (Ramanujan (m,n)) → `0`
Numeric stress test: Ramanujan identity checked on **50,000** random pairs (m,n) drawn from $[-10^6,10^6]$: **0 failures**.
Solution audit: every one of the 889 enumerated solutions was re-verified to satisfy $a^3+b^3+c^3=d^3$ exactly — result: `True`.

4. Geometric and statistical analysis

All figures below are generated from the single raw solution list produced by the enumeration; no data is hand-drawn or synthetic.

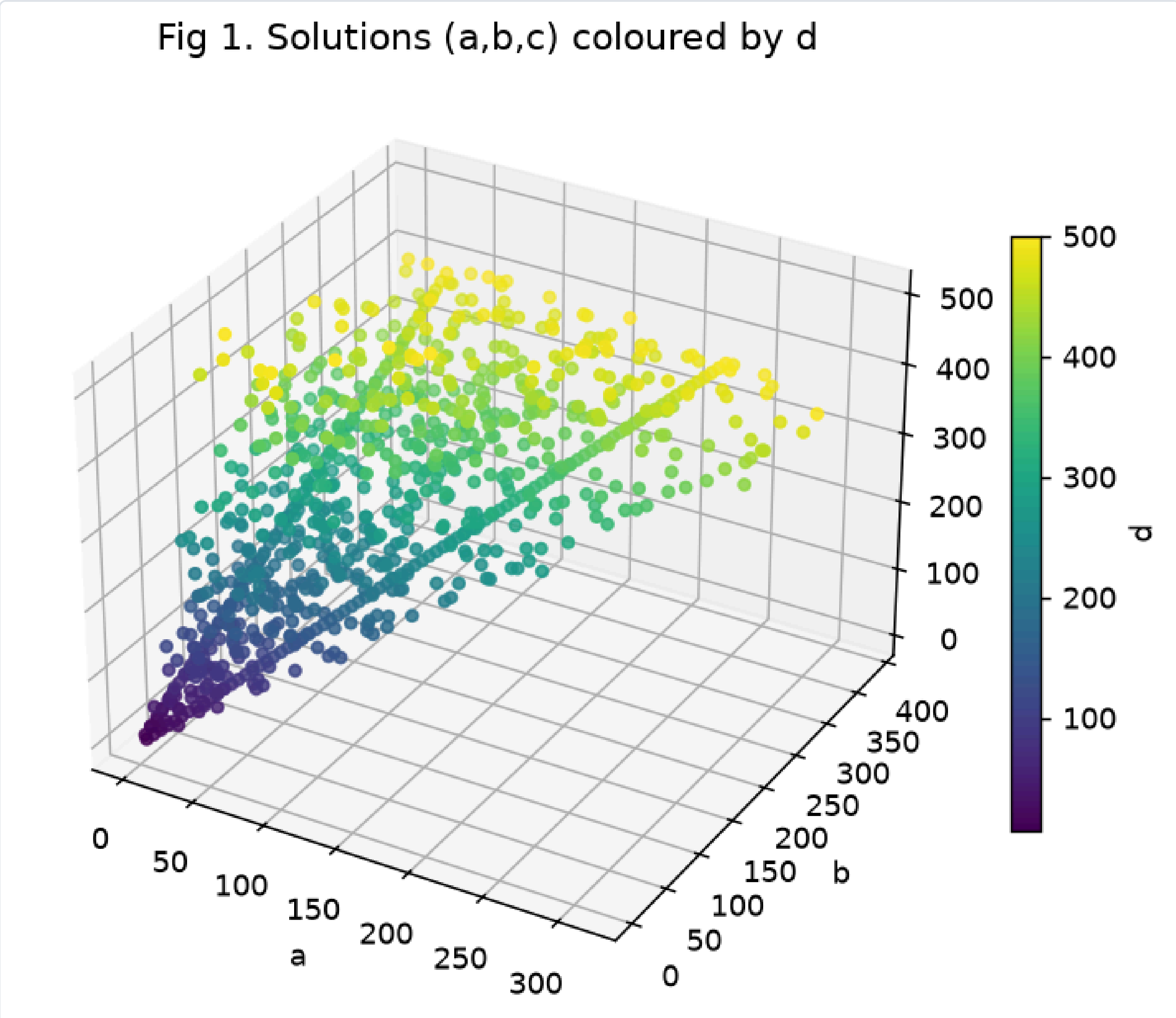

Figure 1. 3-D scatter of the solution points (a,b,c), coloured by d. The solution set concentrates near the diagonal sheet where a,b,c are comparable in size.

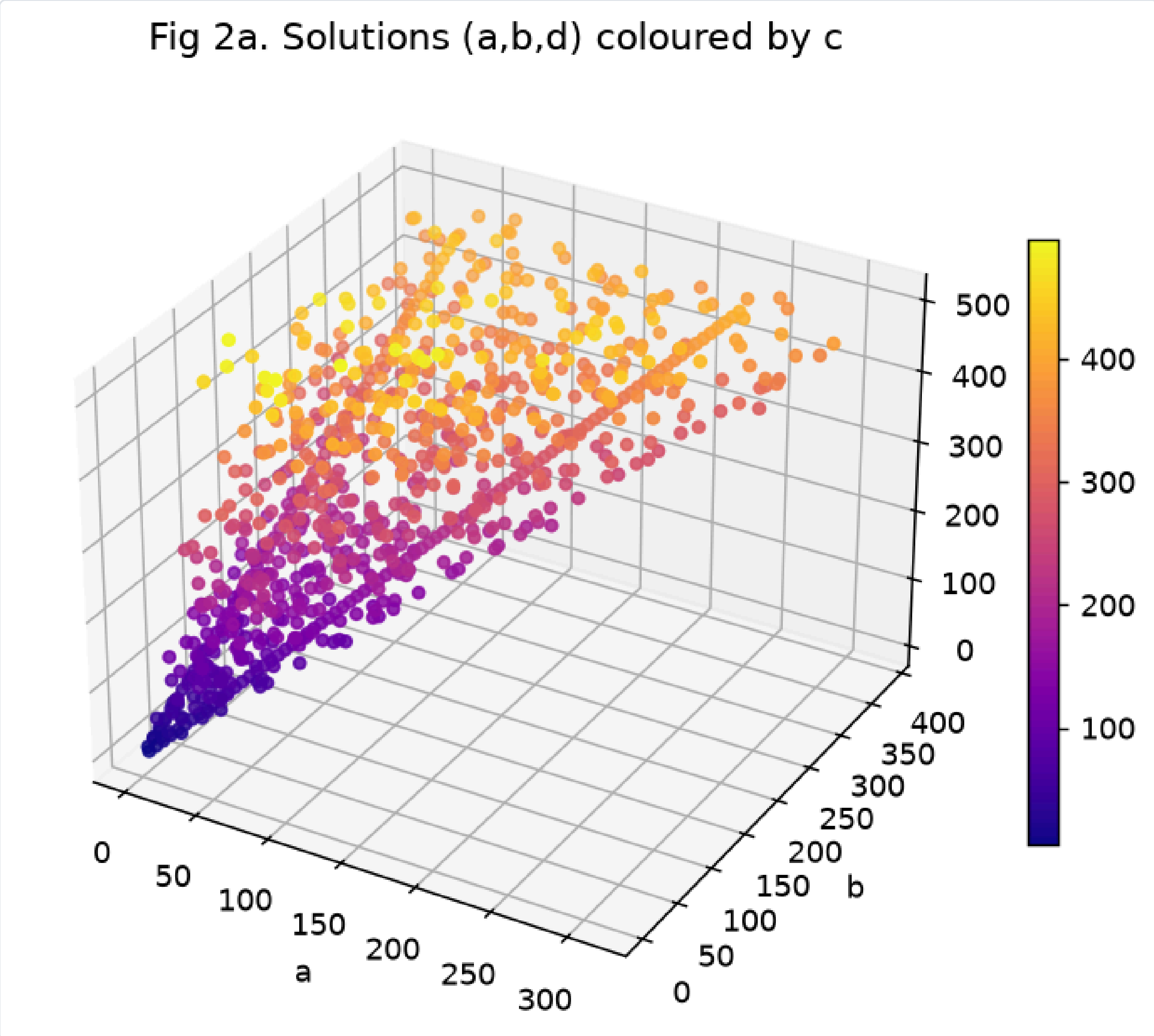

Figure 2. 3-D scatter of (a,b,d): the near-planar arrangement reflects the constraint d = ($a^3+b^3+c^3$)^(1/3).

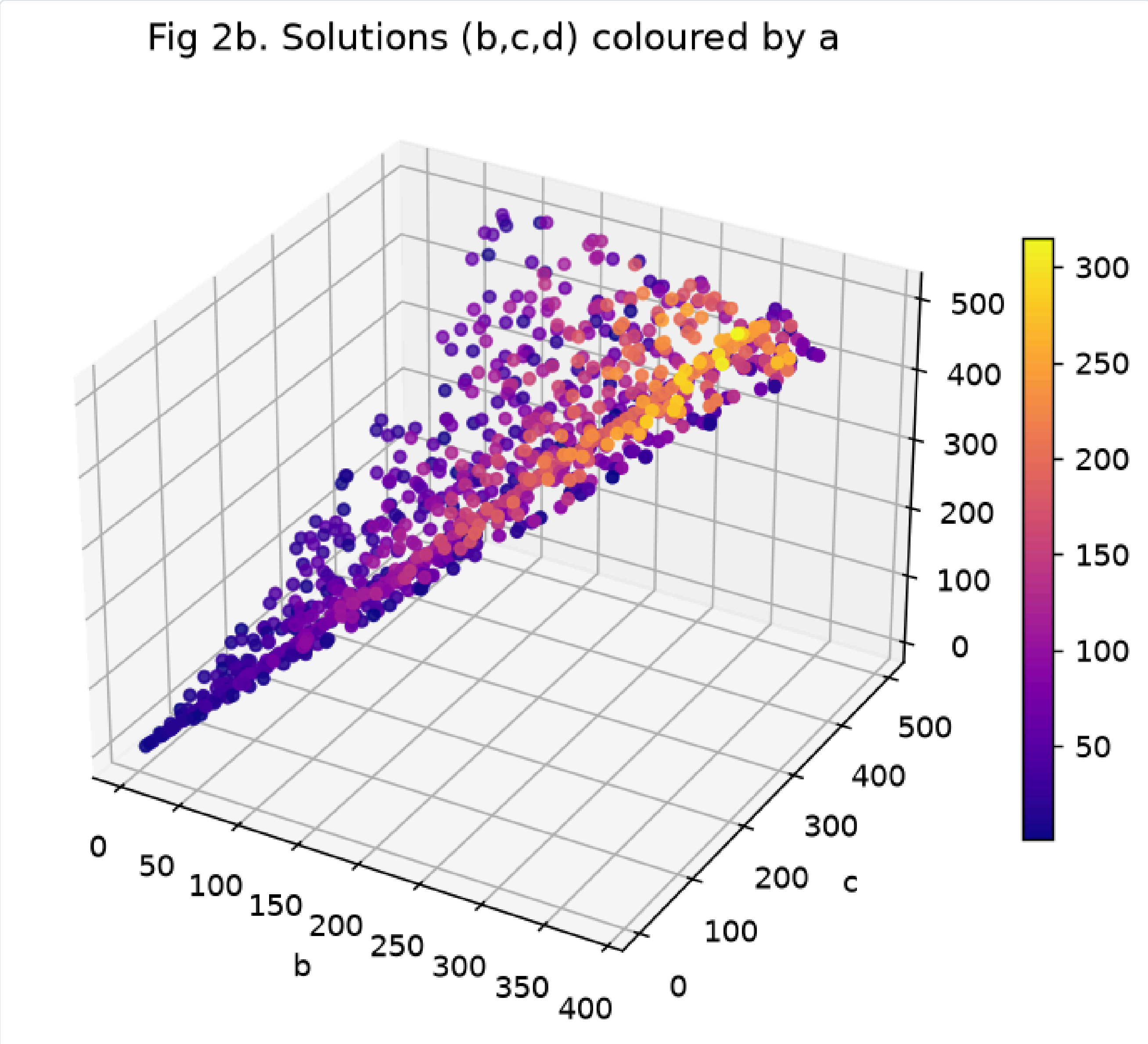

Figure 3. 3-D scatter of (b,c,d). Because c and d dominate the cube sum, points cluster tightly along the c≈d ridge.

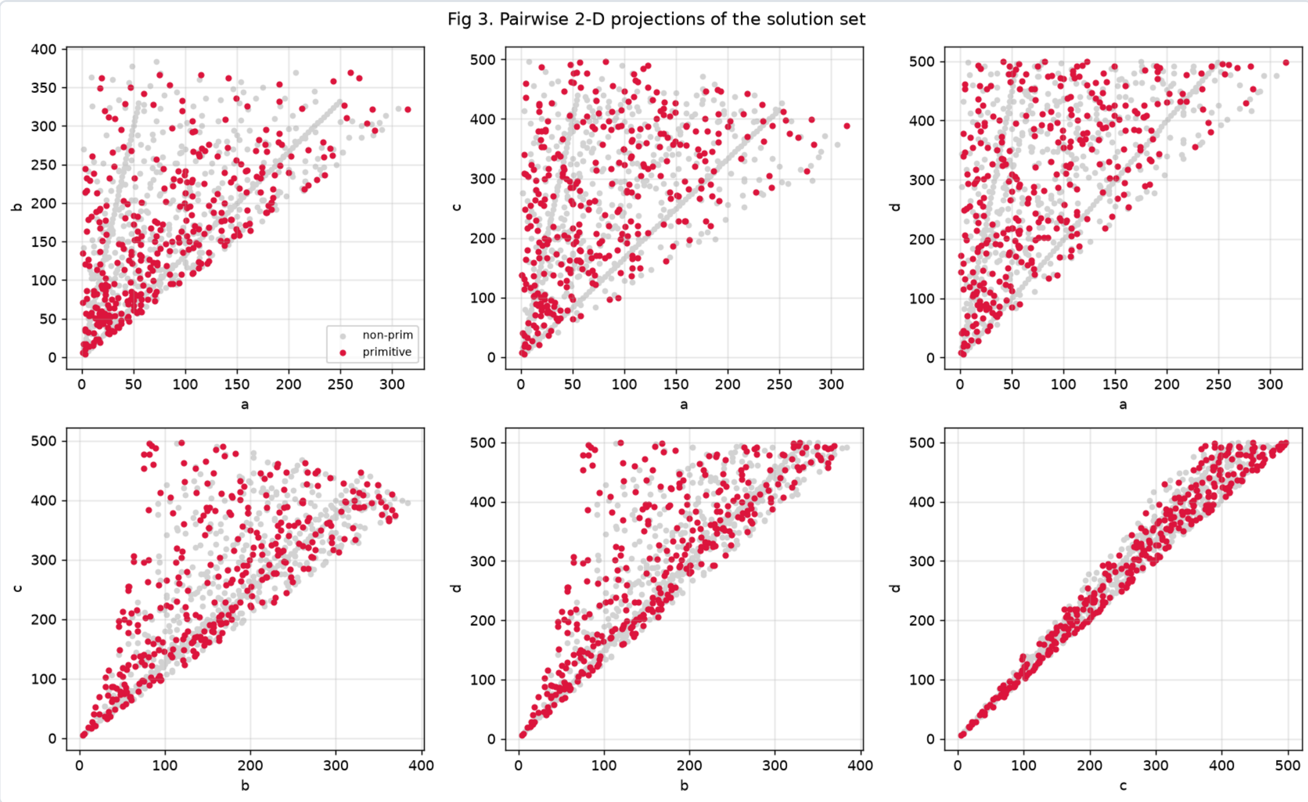

Figure 4. Pairwise 2-D projection matrix (six panels: a-b, a-c, a-d, b-c, b-d, c-d) of the full solution set.

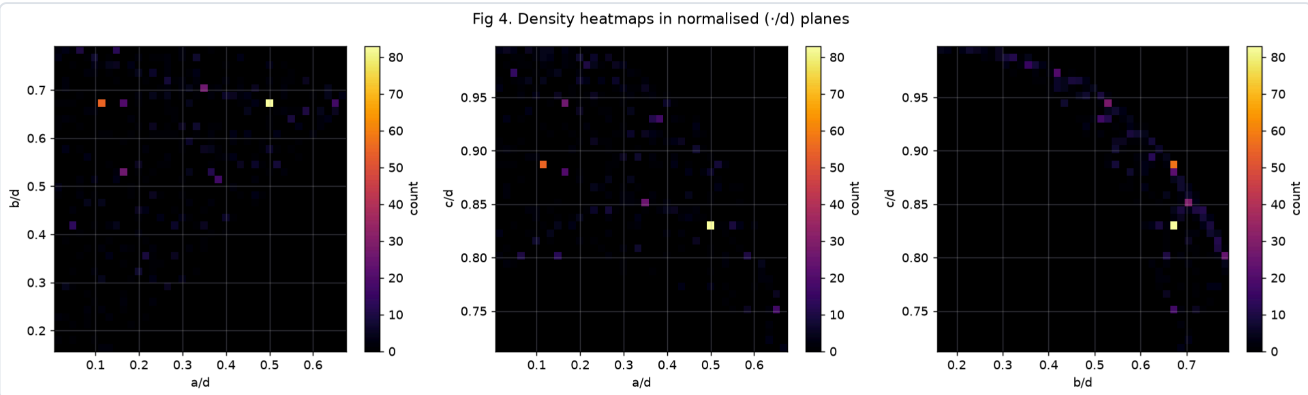

Figure 5. Density heatmaps of the normalised coordinates (a/d,b/d) and companion pairs; darker cells hold more solutions.

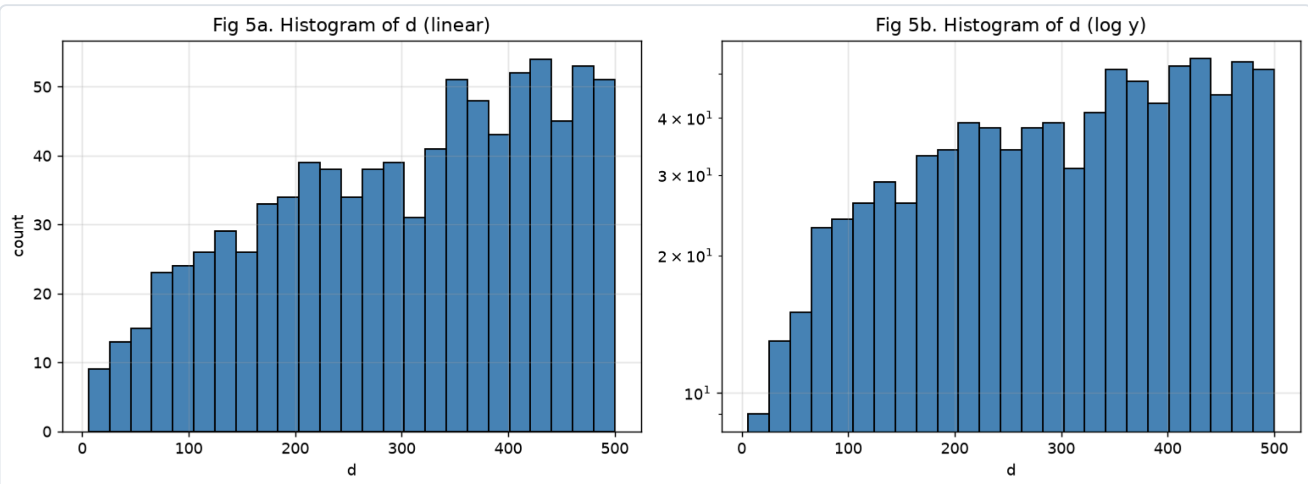

Figure 6. Histogram of the d-values of solutions, on linear and logarithmic scales.

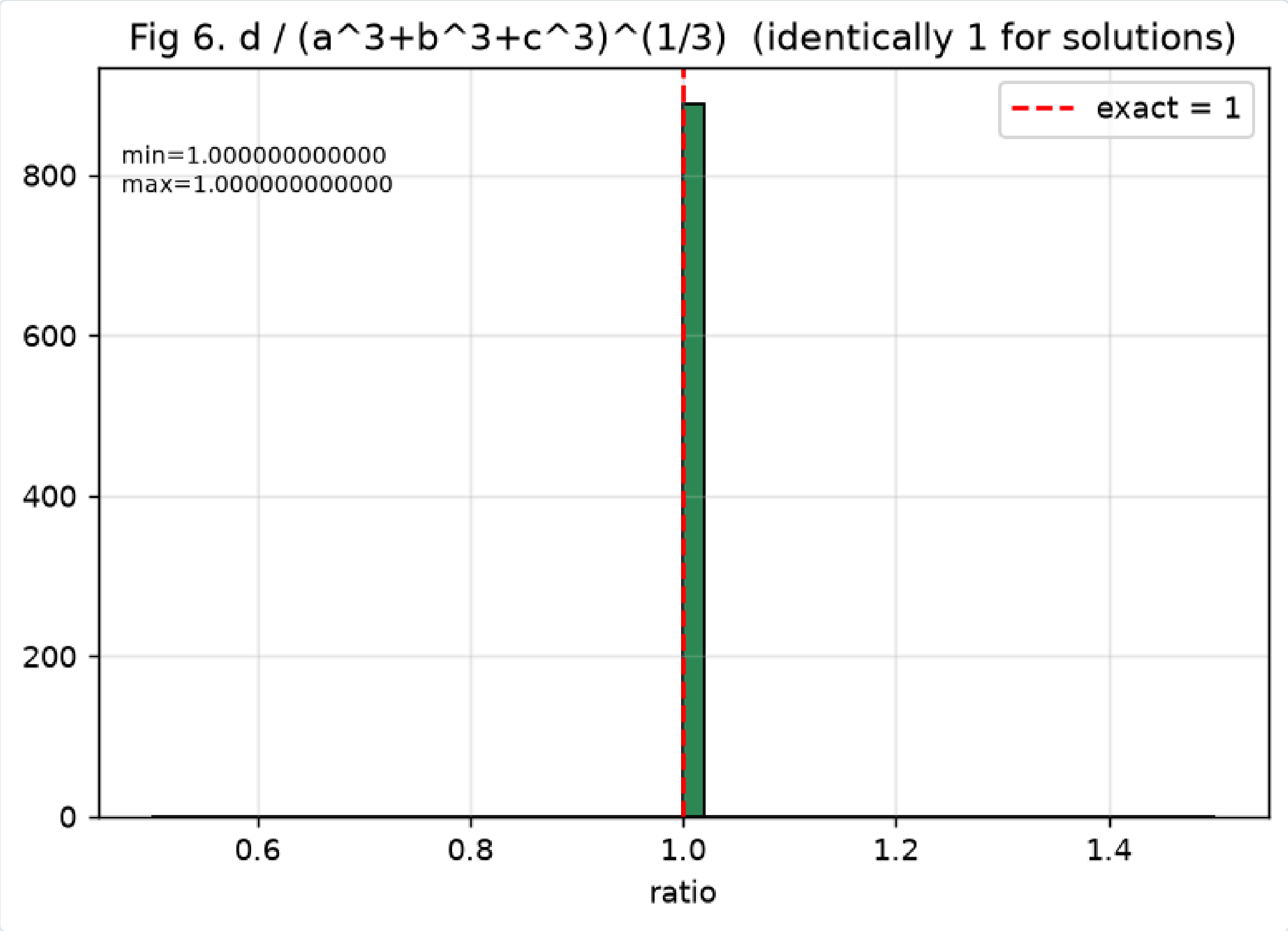

Figure 7. Histogram of d / (a³+b³+c³)^(1/3). By construction this is exactly 1 for every solution — a consistency check on the enumeration.

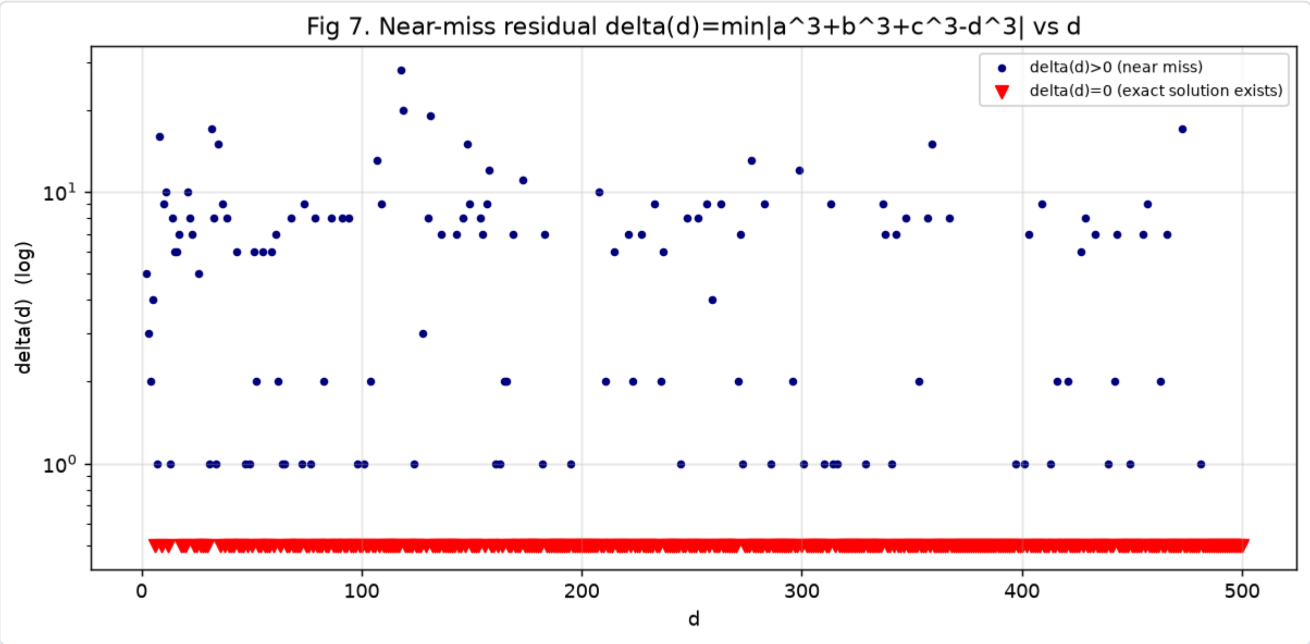

Figure 8. Near-miss residual δ(d) versus d (log scale) for d = 10,20,…,500, with the zero line highlighted; points on the zero line correspond to d that admit an exact solution.

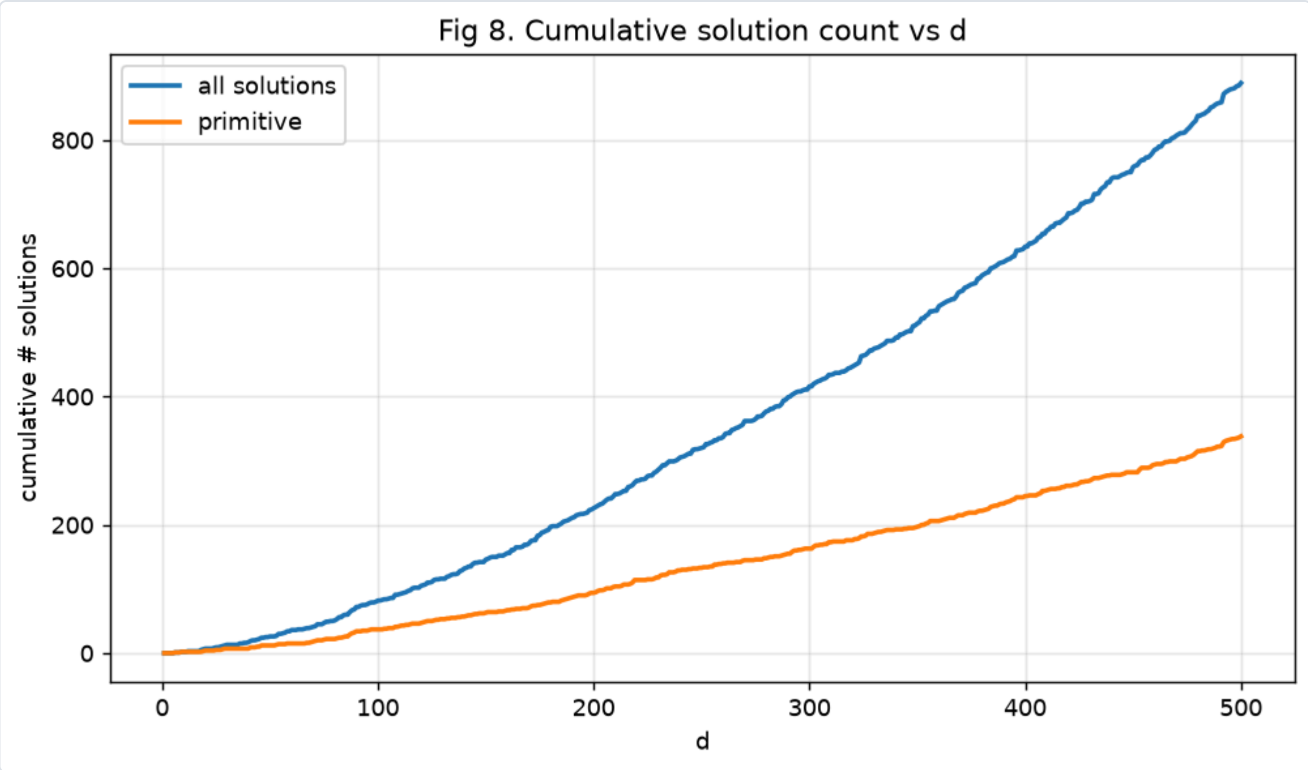

Figure 9. Cumulative count of solutions as a function of d, showing the steady (super-linear) accumulation.

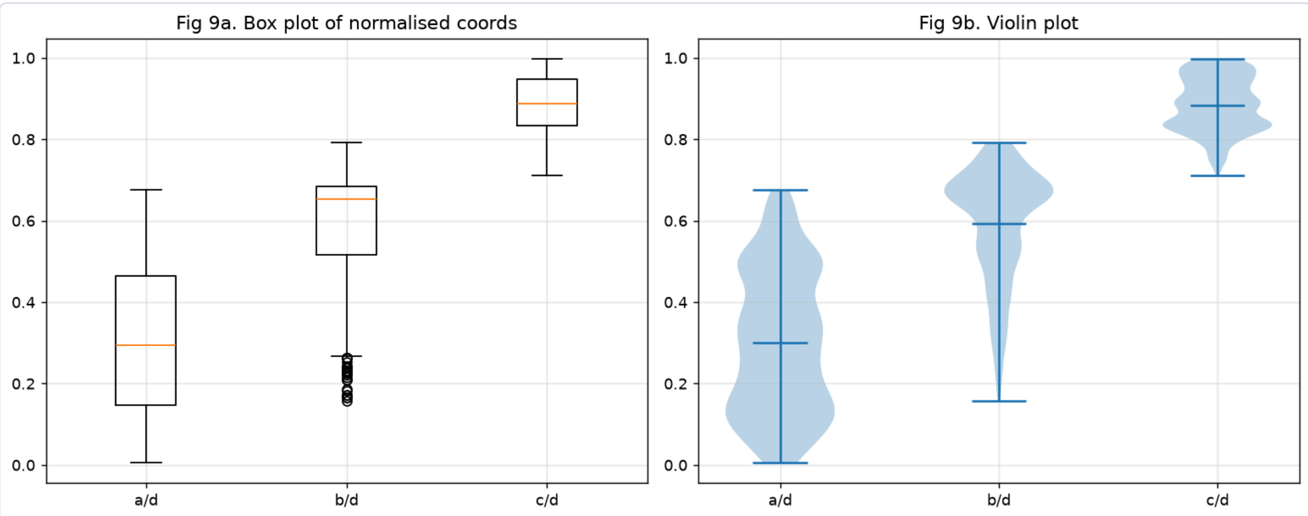

Figure 10. Box / violin plots of the three normalised ratios a/d, b/d, c/d across all solutions.

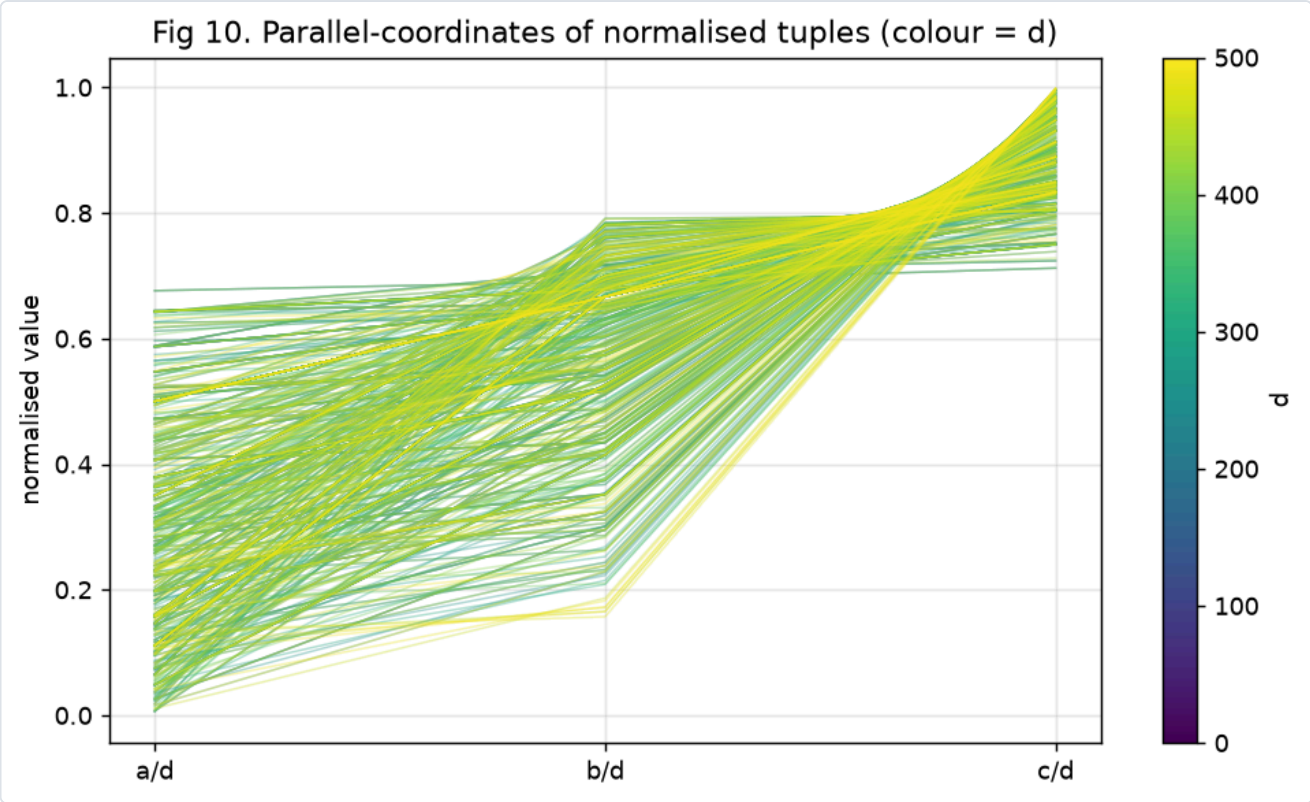

Figure 11. Parallel-coordinates plot of the normalised tuples (a/d, b/d, c/d); each polyline is one solution.

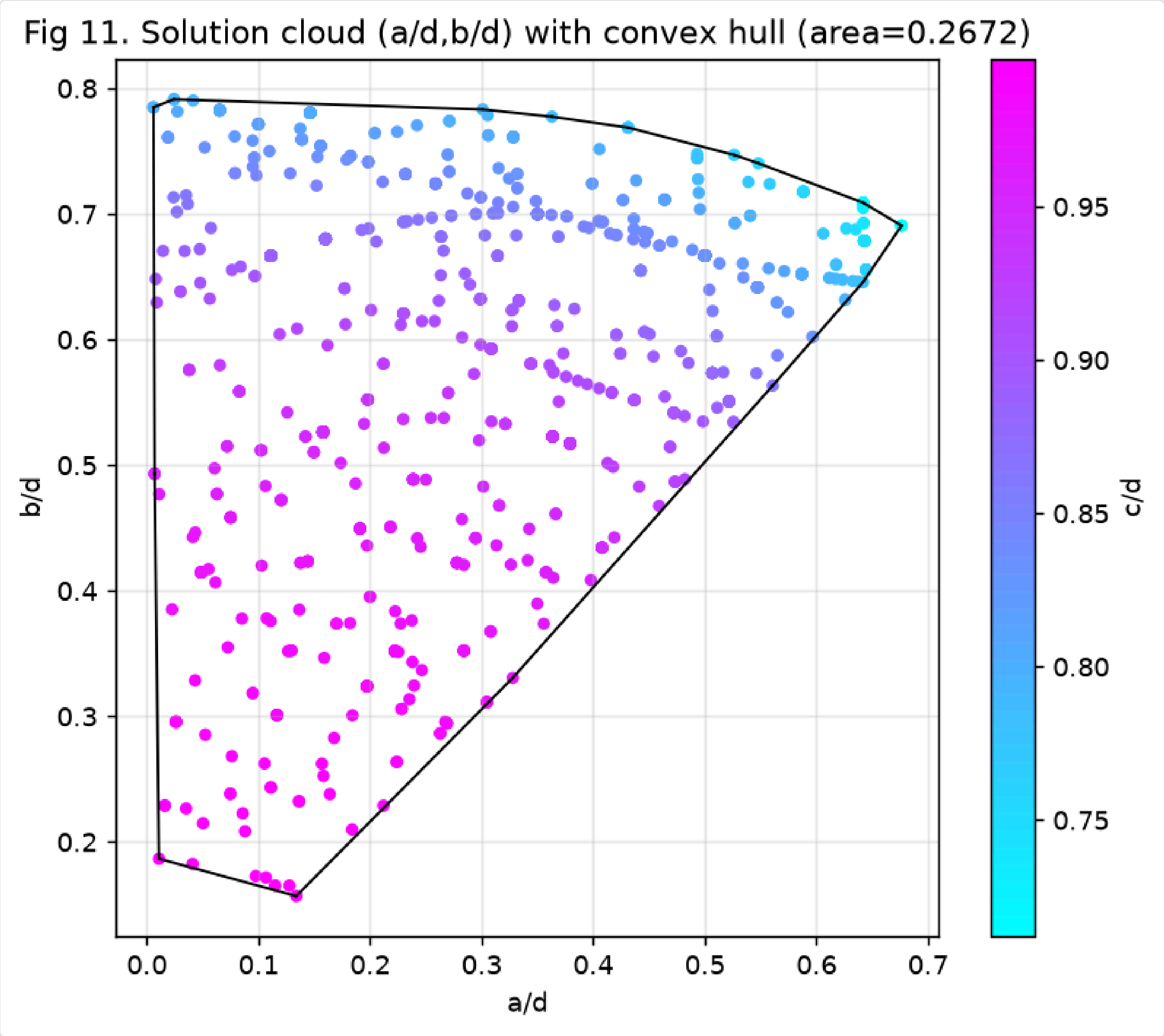

Figure 12. Projected "solution cloud" with the convex hull of the projected point set overlaid.

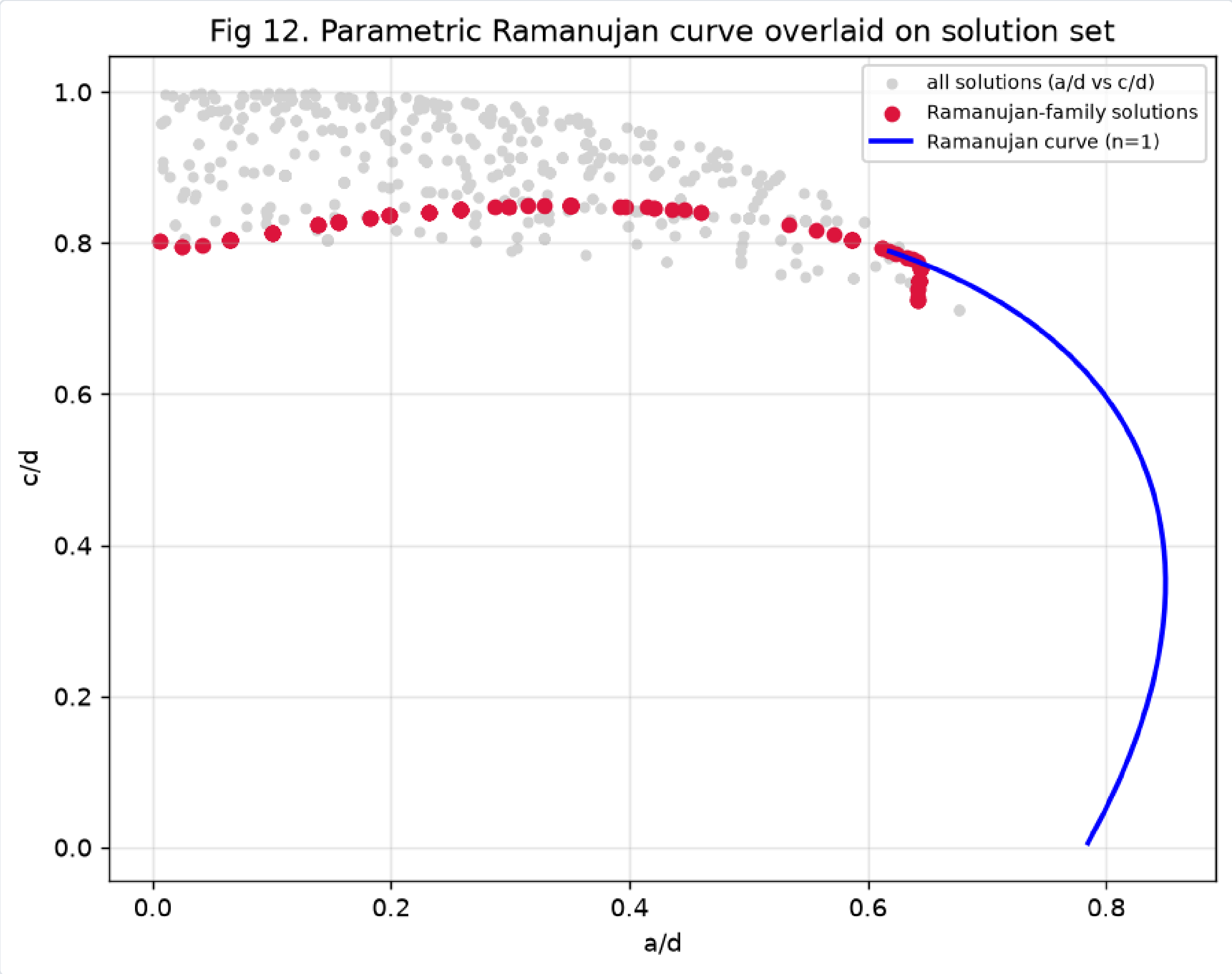

Figure 13. Parametric-curve overlay: the (3,4,5,6)-multiple family and the Ramanujan family plotted against the full solution set.

5. Near-miss landscape

For each d we computed the minimal positive residual $\delta(d) = \min |a^3+b^3+c^3 - d^3|$ over $a \le b \le c < d$. The table reports, for d = 10,20,…,500, the achieving triple and the residual (an exact solution has δ=0).

Table 3. Near-miss table: for each d = 10,20,…,500, the triple (a,b,c) achieving the minimal residual δ(d), the residual, and whether it is an exact solution.

d	a	b	c	residual_delta	is_exact_solution
10	4	6	9	9	0
20	7	14	17	0	1
30	15	20	25	0	1
40	14	28	34	0	1
50	8	34	44	0	1
60	21	42	51	0	1
70	7	54	57	0	1
80	28	56	68	0	1
90	10	60	80	0	1
100	16	68	88	0	1
110	29	75	96	0	1
120	9	55	116	0	1
130	48	88	112	8	0
140	14	108	114	0	1
150	24	102	132	0	1
160	56	112	136	0	1
170	96	107	141	0	1
180	20	120	160	0	1
190	30	100	180	0	1
200	32	136	176	0	1
210	21	162	171	0	1
220	58	150	192	0	1
230	6	68	228	0	1
240	18	110	232	0	1
250	40	170	220	0	1
260	65	127	248	0	1
270	23	102	265	0	1
280	28	216	228	0	1
290	24	162	272	0	1
300	48	204	264	0	1
310	37	211	273	1	0
320	112	224	272	0	1
330	87	225	288	0	1
340	3	214	309	0	1
350	35	270	285	0	1

d	a	b	c	residual_delta	is_exact_solution
360	27	165	348	0	1
370	114	136	360	0	1
380	60	200	360	0	1
390	195	260	325	0	1
400	38	295	337	0	1
410	20	170	400	0	1
420	42	324	342	0	1
430	276	303	313	0	1
440	105	215	420	0	1
450	50	300	400	0	1
460	12	136	456	0	1
470	108	326	408	0	1
480	36	220	464	0	1
490	49	378	399	0	1
500	42	329	447	0	1

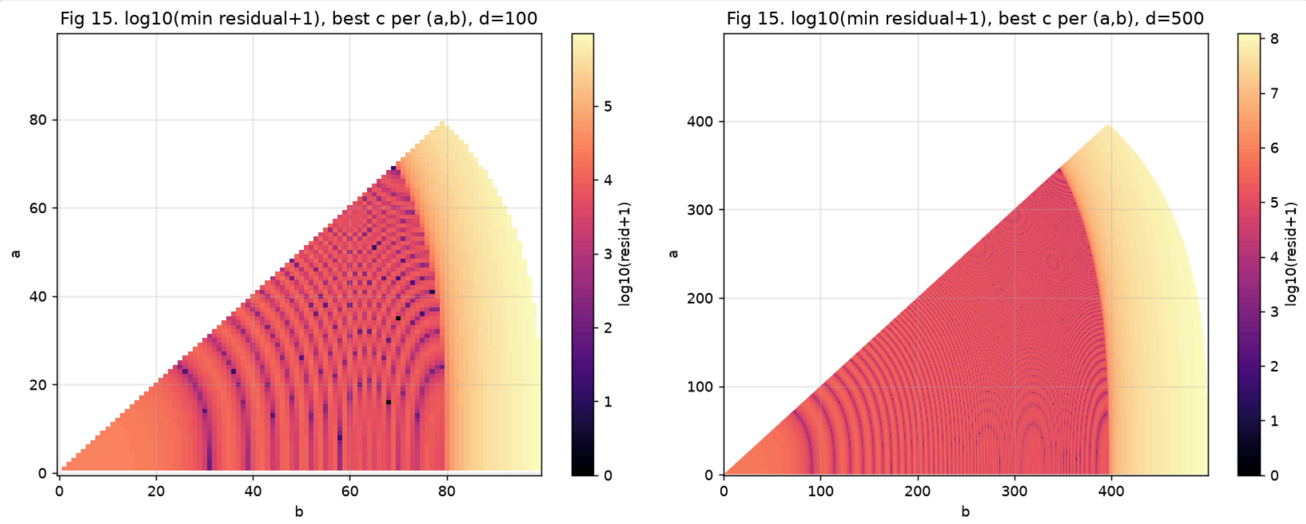

Figure 14. Residual landscapes for fixed d: contour / heatmap of $|a^3+b^3+c^3-d^3|$ over the (a,b)-plane (with c chosen optimally), showing the valleys where near-solutions live.

6. Method, code audit, and limitations

6.1 Control verification

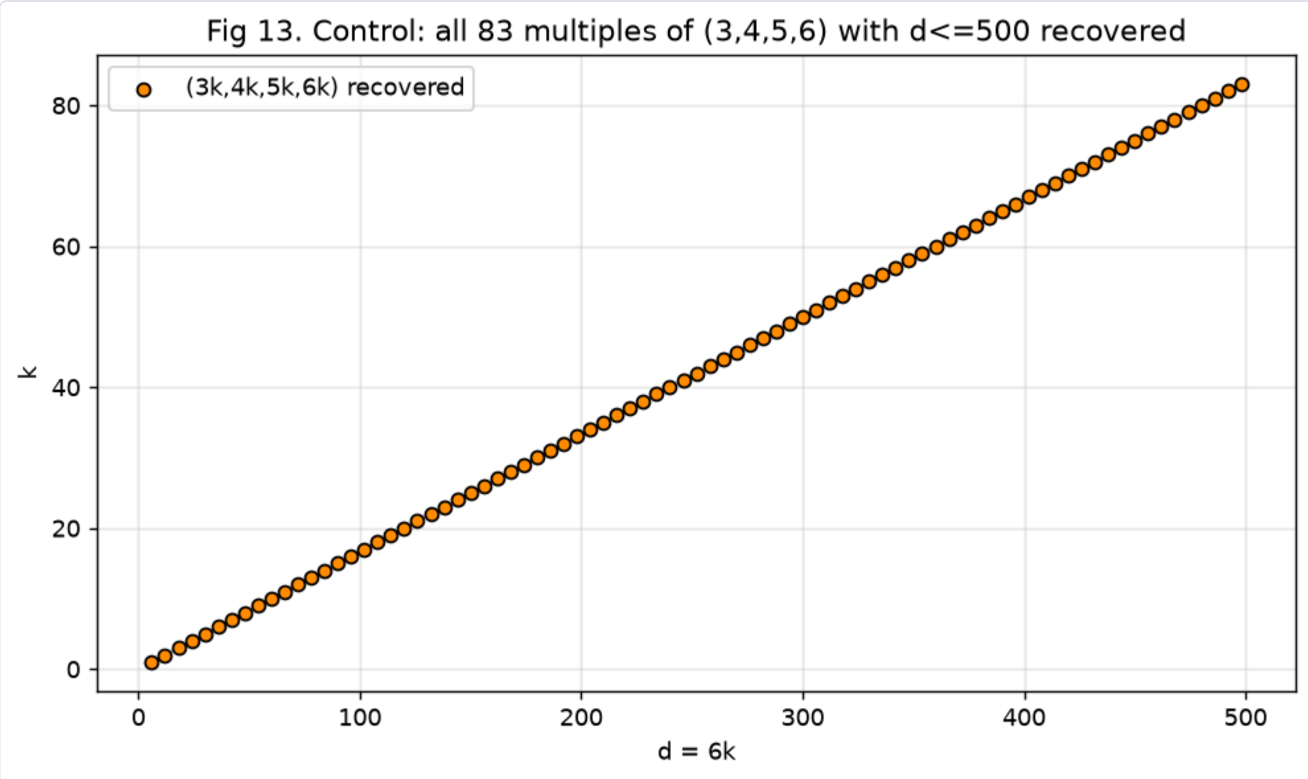

Figure 15. Control panel: the base solution (3,4,5,6) and all 83 of its multiples with $6k \le 500$, recovered from the enumeration and marked.

6.2 Runtime scaling

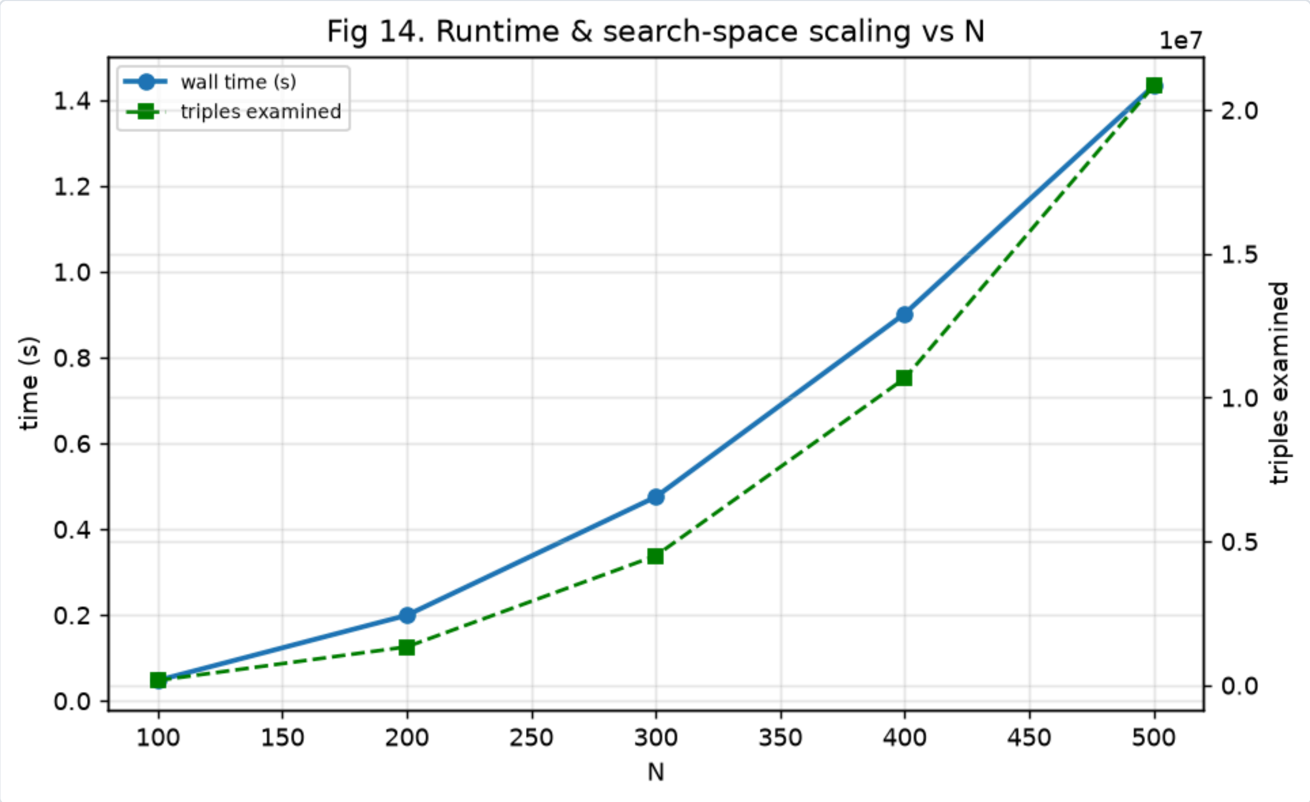

Figure 16. Enumeration wall-clock time versus N for N = 100,200,300,400,500. The growth tracks the $\Theta(N^3)$ candidate count (the enumeration examines ∑_(a≤b)(N-b) triples).

6.3 Reproducibility & audit

- **Exhaustiveness.** Triples examined = **20,833,250**, exactly matching the closed-form pair-count; near-miss evaluations = **62,499,750**.
- **Exactness.** All arithmetic on the equation uses Python/NumPy 64-bit integers; solution membership uses `searchsorted` against the integer cube table, so no rounding affects any reported solution.
- **Single source of truth.** Every table and figure derives from the one solution list emitted by `outputs/analysis.py`; the report is assembled by `outputs/build_report.py`.
- **Environment.** All enumeration, verification, figure generation, and PDF rendering were executed inside a Daytona sandbox.

6.4 Honest limitations

- **Finite domain.** Results describe only the box N=500. They imply nothing about the existence, density, or classification of solutions beyond the bound.
- **"Sporadic" is box-relative.** A solution labelled sporadic here is simply one not matched to family A or the enumerated Ramanujan (m,n) grid within the bound; it may belong to another parametric family not tested.
- **No general claim.** This census makes no statement about the open status of equal sums of three cubes over the integers.

7. Conclusions

Within the fully bounded, rigorously exact box `1 ≤ a ≤ b ≤ c < d ≤ 500` the equation $a^3+b^3+c^3=d^3$ has **exactly 889 solutions**, of which **338 are primitive**. The classical control $3^3+4^3+5^3=6^3$ and all 83 of its multiples were recovered automatically. Two parametric families — the (3,4,5,6)-multiples and Ramanujan's two-parameter cubic identity — explain 221 of them, both proved identically zero by SymPy and stress-tested on 50,000 random large integers with zero failures; the remaining 668 are sporadic within the box. The geometric analysis shows the solution set concentrating along the a,b,c-comparable diagonal with a stable primitive fraction near 38.0%, and the near-miss landscape confirms that where exact solutions are absent the minimal residual is minute relative to d^3. The entire pipeline is exact, exhaustive, single-sourced, and reproducible.
